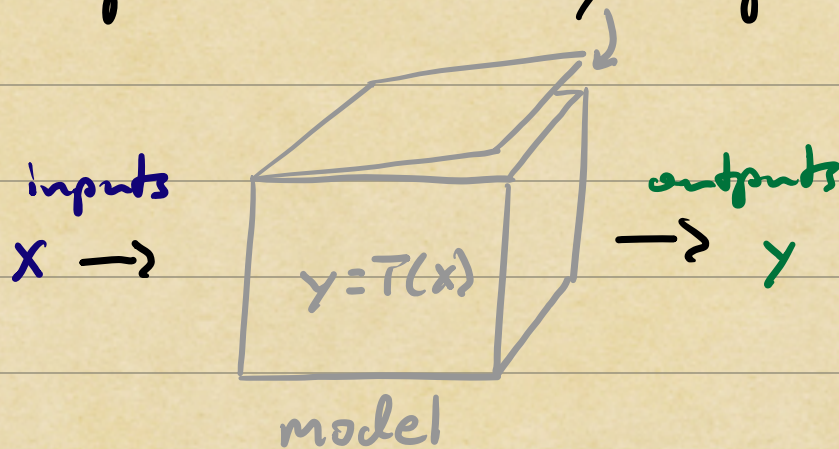


Methods of Applied Math

Fall 2026 Course Info

Linked via LMS github.com/ajhphros23/MATH-6600
Piazza - Q!A, Discussion, Announcements
Gradescope - submit HW+Project, view grades

Aim: Develop toolkit to study complex phenomena.



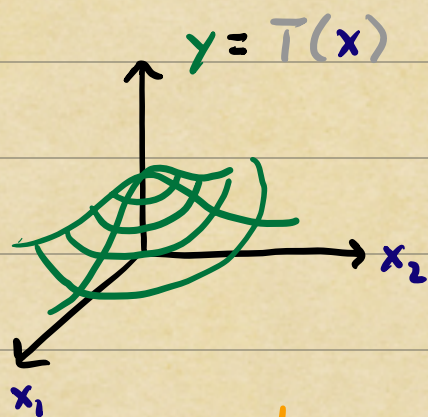
Challenges: In real-world applications, we face

- $\Rightarrow$ High-dimensional inputs/outputs/model
- $\Rightarrow$ Nonlinear coupling of inputs/outputs
- $\Rightarrow$ Partial knowledge of inputs/outputs/model
- $\Rightarrow$ Noise ! corruption in inputs/outputs/model

Often need to work w/computer

Theme: "Slice" x, T, y into "simpler" pieces.

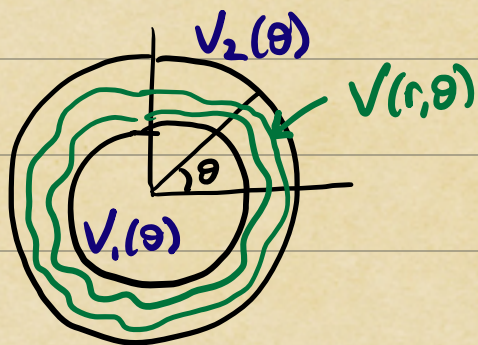
Example 1: "Fit" model to data



Given observations of input/output
 $(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)})$
can we "learn" the map T ?

Linear ; nonlinear approximation

Example 2: Equilibrium Configurations

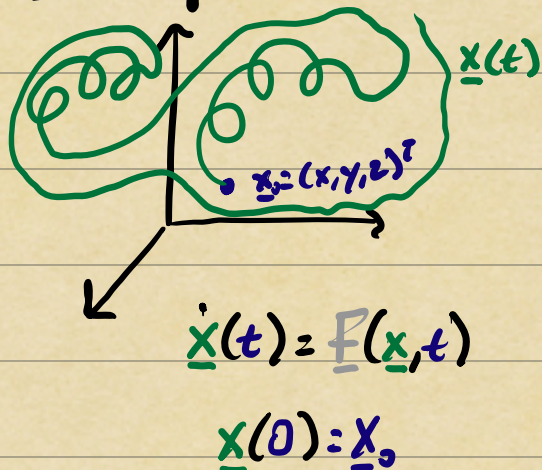


If the inner and outer
surfaces of a hollow dielectric
are held at electric potentials
 $V_1(\theta)$ and $V_2(\theta)$, what is the
potential $V(r, \theta)$ in the tube?

Laplace Eqn, $\Delta V = 0$
+ boundary conditions

Elliptic PDEs (linear) ; Variational Principles (nonlinear)

Example 3: Dynamical Systems



Given initial condition $\underline{x}_0$
and dynamical law $\underline{F}$,
how does $\underline{x}(t)$ evolve?

$$\dot{\underline{x}}(t) = \underline{F}(\underline{x}, t)$$

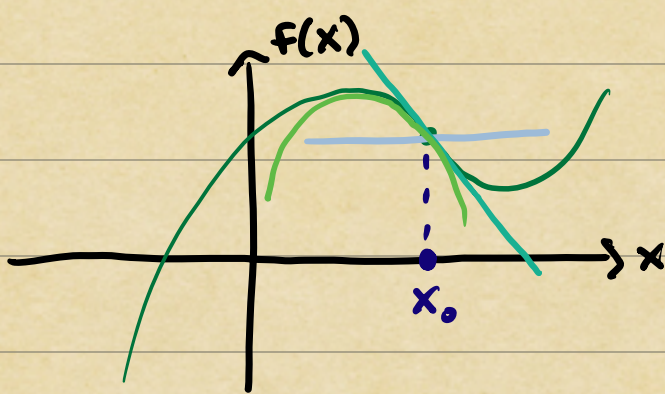
$$\underline{x}(0) = \underline{x}_0$$

Lin. PDEs (linear inf-dim), Nonlinear Dynamics, Nonlinear PDE

2 Familiar Ways to "Slice" Functions

One of the most powerful ideas in applied mathematics is that complicated functions can be built up by combining simpler ones.

Example 1: Taylor Polynomials

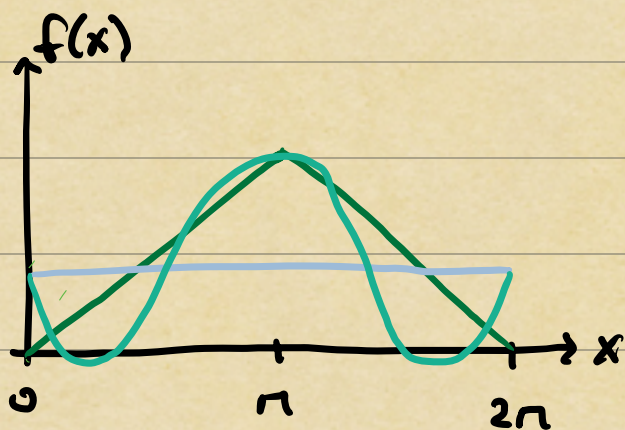


$$\begin{aligned} f(x) &\approx f(x_0) + f'(x_0)(x-x_0) \\ &\quad + \frac{1}{2} f''(x_0)(x-x_0)^2 \\ &\quad \vdots \\ &\quad + \frac{1}{n!} f^{(n)}(x_0)(x-x_0)^n \end{aligned}$$

Build up $f(x)$ near x_0 using monomials $(x-x_0)^j$.

- $\Rightarrow$ Polynomials are easy to calculate with!
- $\Rightarrow$ Consequently, poly. approx. are at the heart of many analytic and numerical techniques.
- $\Rightarrow$ Series soln's for ODEs, rootfinders, optimizers, finite difference/element/volume methods.

Example 2: Fourier Series



$$f(x) \approx \frac{a_0}{2} + a_1 \cos(x)$$

$$+ a_2 \cos(2x)$$

$\vdots$

$$+ a_n \cos(nx)$$

↑
discuss coeffs later

Build up $f(x)$ using trigonometric functions.

$\Rightarrow$ Trig. functions are also easy to work with!

$\Rightarrow$ Trig. approx. is central to many theoretical and computational tasks in signal processing and the solution/analysis of ODE/PDE.

$\Rightarrow$ Image processing, Denoising, Spectral filtering, Stability analysis, wave prop.

Q: What features do Taylor and Fourier series have in common?

Q: In which features do Taylor and Fourier series differ?